



Planning of a charging station for electric and hydrogen vehicles under hydrogen storage and fuel cell systems using a novel stochastic p-robust optimization technique

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ABSTRACT

This article presented a robust plan for an off-grid charging station (OGCS) for electric vehicles (EVs) and hydrogen vehicles (HVs) based on a photovoltaic (PV) system and a hydrogen storage system (HSS). This OGCS simultaneously supplies HVs and EVs continuously throughout the day. Also, HSS and fuel cell (FC) systems have been allocated in the OGCS to be used when we do not have access to the power of the PV system. In addition, a diesel generator (DG) is also designed to prevent in cases where we have extreme uncertainty, including the lack of energy in the PV system and the high load of the system, which may lead to load interruption. Uncertainties of electric and hydrogen loads of EVs and HVs in addition to PV production power are simulated using scenario-based stochastic optimization technique (SOT). Finally, a new framework based on stochastic p-robust optimization technique (SPROT) is applied to optimize the maximum relative regret (MRR) in the worst scenario in order to achieve robust planning in the uncertain environment. The obtained results from the proposed SPROT are compared with SOT. The compared results indicate a 4.51% raise in the average cost in SPROT and a 45.73% decrease in MRR that leads to robust planning. Finally, installed capacity of PV system will decrease from 1688 to 1685 kW, while installed capacity of DG will increase from 78 to 123 kW.

1. Introduction

Today, renewable energy units such as PV, FC, and HSS are increasing, which leads to different uses of these sources [1]. One of the important applications of these sources is the supply of energy for EVs and HVs in OGCS separated from the national power grid [2]. These charging stations need a storage system to store excess energy from renewable energy sources when necessary and use it when the primary energy source is unavailable [3]. These storage systems can be batteries, hydrogen storage systems [4], or other technologies used [5].

1.1. Literature review

In modern and sustainable societies and cities, it is necessary to create OGCS for EVs and HVs [6]. FC-based HVs work by directly converting hydrogen into electricity, which is environmentally friendly

compared to traditional cars [7]. Also, these cars have little pollution and low noise, which can be a good substitute for gasoline and diesel cars in the future [8]. Therefore, it is necessary to design a charging station for electric and hydrogen vehicles that do not have access to the national grid and use primary energy from renewable sources [9].

Charging stations are classified into grid-connected and off-grid modes.

The following articles can be presented in the grid-connected mode. In Ref. [10], the energy purchased cost of a smart parking lot is minimized in the presence of HSS, which uncertainty parameter is modeled via stochastic programming, and the PSO algorithm is applied to solve this problem. The IGDT method is applied to model the uncertainty of market price in order to obtain the bidding curve of an HSS-based parking lot in Ref. [11]. In Ref. [12], the rated power of battery storage and solar panels is determined via SOT. A technique based on adaptive learning is presented in Ref. [13] for EVs charging that the uncertainties of solar energy and electricity price are modeled via

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